Java Programming: Spacecraft Fleet Management

Implement a set of classes to manage spacecraft fleets using Java's cloning mechanism and static tracking. The task includes implementing detailed functionality for engine configurations, abstract spacecraft, concrete shuttle/probe classes, and a control center to demonstrate operations.

Class Specifications:

(1) **EngineConfig** class (Implements Cloneable interface)

Fields:

- private String engineID: Engine ID.

Methods:

- public EngineConfig(String engineID): Constructor.
- public EngineConfig clone(): Overrides clone() to return a deep copy of EngineConfig.
- public void setEngineID(String engineID): Mutator method for the field engineID.
- public String toString(): Returns the String representing an EngineConfig object.

(2) Abstract **Spaceship** class (Implements Cloneable interface)

Fields:

- private final double fuel: Fixed fuel quantity (immutable).
- private EngineConfig engine: Engine configuration.
- private static int totalInstances: Total instances created (including clones).

Methods:

- public Spaceship(double fuel, String engineID): Constructor (Initializes all fields and increments totalInstances).
- public static int getTotalInstances(): Accessor method for the static field totalInstances.
- public double getFuel(): Accessor method for the field fuel.
- public void setEngine(EngineConfig engine): Mutator method for the field engine.
- public EngineConfig getEngine(): Accessor method for the field engine.
- public Spaceship clone(): Overrides clone() to create deep copy of engine field via engine.clone(). Increments totalInstances. Returns cloned Spaceship.

(3) **CargoShuttle** class (Inheriting from Spaceship)

Fields:

- private double cargoCapacity: Current cargo load.

Methods:

- public CargoShuttle(double fuel, String engineID, double cargoCapacity): Constructor: initializes fuel and engineID by invoking super class's constructor, then initializes cargoCapacity.
- public void loadCargo(double amount): Adds amount to cargoCapacity.
- public void unloadCargo(): Resets cargoCapacity to 0.
- public double getCargoCapacity(): Accessor method for the field cargoCapacity.
- public CargoShuttle clone(): Covariant override, calling super class's clone() method.
- public String toString(): Returns the String representing an CargoShuttle object (including fuel, engine, and cargoCapacity fields).

(4) **ExplorerProbe** class (Inheriting from Spaceship)

Fields:

- private ArrayList<String> scientificInstruments: List of instruments.

Methods:

- public ExplorerProbe(double fuel, String engineID, String... instruments): Constructor: initializes fuel and engine fields by invoking super class's constructor, then appends instrument to scientificInstruments.
- public void addInstrument(String instrument): Appends instrument to scientificInstruments.
- public ExplorerProbe clone(): Covariant override; creates new ArrayList copy of scientificInstruments.
- public String toString(): Returns the String representing an ExplorerProbe object (including fuel, engine, and scientificInstruments).

(5) **SpaceControlCenter** class (Main Class)

The main method simulates fleet operations:

- Create an empty ArrayList named fleet as the container of Spaceship objects.
- Add two objects to fleet:

CargoShuttle: fuel=5000, engineID="CT-001", cargoCapacity=10000

ExplorerProbe: fuel=300, engineID="EP-ALPHA", instruments ["Spectrometer", "Camera"]

- Clone the first element (CargoShuttle) and add to fleet.
- For cloned CargoShuttle (index 2):
  - Load 500 cargo.
  - Change engine ID to "CT-COPY".
- For ExplorerProbe (index 1):
  - Add instrument "Laser Drill" via addInstrument().
- Sort fleet by ascending fuel using: Collections.sort();
- Print each spacecraft in fleet.
- Print total spacecraft number.

Expected Output:

```
ExplorerProbe [fuel=300.0, engine=EngineConfig [engineID=EP-ALPHA], scientificInstruments=[Spectrometer, Camera, Laser Drill]]
CargoShuttle [fuel=5000.0, engine=EngineConfig [engineID=CT-001], cargoCapacity=10000.0]
CargoShuttle [fuel=5000.0, engine=EngineConfig [engineID=CT-COPY], cargoCapacity=10500.0]
Total spacecraft: 3
```